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INTRODUCTION TO SOFTWARE ENGINEERING

ASSIGNMET#1



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ASSIGNMENT 1

LEARNING MANAGEMENT SYSTEM

The Learning Management System (LMS) is a web and mobile-based platform designed to facilitate online learning and course management. It provides students and instructors with tools for course enrollment, progress tracking, attendance marking, assessments, and course recommendations.

Key Features:

1. Course Management

- Instructors can create, upload, and manage course materials (lecture notes, quizzes, assignments).
- Students can enroll in and drop courses dynamically.
- The system recommends courses based on past enrollments and interest grades.

2. Student Progress & Evaluation

- Automated grading of quizzes and assignments.
- Student progress tracking with real-time grade updates.
- Students and instructors can view detailed performance reports.

3. Attendance Management

- Instructors manually mark student attendance for each class session.
- Attendance records are stored and updated in real time.
- Students can view their attendance reports.

4. Feedback & Reviews

- Students can rate courses and provide reviews after completion.
- The system stores and displays course feedback to help improve course quality.

5. Security & Compliance

- All user data is encrypted using AES-256 to ensure security.
- The system ensures 99.9% uptime for uninterrupted access.
- Complies with GDPR & FERPA regulations to protect user privacy.

6. User Roles & Access Control

- Admin: Manages users, courses, and system settings.
- Instructors: Upload course materials, grade students, and mark attendance.
- Students: Enroll in courses, complete assignments, and track progress.

A-FUNCTIONAL REQUIREMENTS

1. Data Requirements

1. The system should store user profiles, including name, email, and role (student/instructor/admin).
2. Course materials (videos, PDFs, quizzes) should be uploaded, stored, and retrieved by students.
3. Student progress and grades should be automatically updated after completing quizzes or assignments.
4. The system should allow instructors to export reports on student performance in CSV/PDF formats.

REQUIREMENT MANAGEMENT TOOL

JIRA DATA REQUIREMENTS

FR01: The system should store user profiles, including name, email, and role (student/instructor/admin).	DATA REQUIREMENTS
FR02: Course materials (videos, PDFs, quizzes) should be uploaded, stored, and retrieved by students.	DATA REQUIREMENTS
FR03: Student progress and grades should be automatically updated after completing quizzes or assi...	DATA REQUIREMENTS
FR04: The system should allow instructors to export reports on student performance in CSV/PDF for...	DATA REQUIREMENTS

2. Interface Requirements

1. The LMS should have a dashboard for students, instructors, and admins with relevant stats.
2. The interface should support dark mode and light mode for better user experience.
3. Course content should be displayed in a structured format with a navigation bar.
4. The system should provide a search bar to allow users to quickly find courses, materials, or discussions.

REQUIREMENT MANAGEMENT TOOL

JIRA INTERFACE REQUIREMENTS

FR05: The LMS should have a dashboard for students, instructors, and admins with relevant stats.	INTERFACE REQUIREM...
FR06: The interface should support dark mode and light mode for better user experience.	INTERFACE REQUIREM...
FR07: Course content should be displayed in a structured format with a navigation bar.	INTERFACE REQUIREM...
FR08: The system should provide a search bar to allow users to quickly find courses, materials, or dis...	INTERFACE REQUIREM...

3. Personalization Requirements

1. Users should be able to customize their profiles, including profile pictures and bios.

2. The system should recommend courses based on past enrollments and interests.
3. Students should be able to set learning goals and receive progress reminders.
4. Users should be able to customize notification settings (email, SMS, or in-app alerts).

REQUIREMENT MANAGEMENT TOOL

JIRA PERSONALIZATION REQUIREMENTS

FR09: Users should be able to customize their profiles, including profile pictures and bios.

PERSONALIZATION RE...

FR10: The system should recommend courses based on past enrollments and interests.

PERSONALIZATION RE...

FR11: Students should be able to set learning goals and receive progress reminders.

PERSONALIZATION RE...

FR12: Users should be able to customize notification settings (email, SMS, or in-app alerts).

PERSONALIZATION RE...

4. Transactional Requirements

1. Students should be able to enroll in and drop courses dynamically.
 2. Instructors should be able to create, update, and delete courses.
 3. The system should support secure payments for premium courses using PayPal or credit cards.
 4. A feedback system should allow students to rate courses and provide reviews after completion.
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REQUIREMENT MANAGEMENT TOOL

FR13: Students should be able to enroll in and drop courses dynamically.

TRANSACTIONAL REQ...

FR14: Instructors should be able to create, update, and delete courses.

TRANSACTIONAL REQ...

FR15: The system should support secure payments for premium courses using PayPal or credit cards.

TRANSACTIONAL REQ...

FR16: A feedback system should allow students to rate courses and provide reviews after completion.

TRANSACTIONAL REQ...

B-NON FUNCTIONAL REQUIREMENTS

1. Expected Quality Requirements

1. The system should be able to handle at least 500 concurrent users without lag.
2. Page load time should not exceed 2 seconds under normal conditions.
3. The system should ensure 99.9% uptime for uninterrupted access.
4. Data should be encrypted using AES-256 to ensure security of user information.

REQUIREMENT MANAGEMENT TOOL

JIRA Expected Quality Requirements

NFR01: The system should be able to handle at least 500 concurrent users without any lag.	QUALITY REQUIREMEN...
NFR02: Page load time should not exceed 2 seconds under normal conditions.	QUALITY REQUIREMEN...
NFR03: The system should be available and operational 99.9% of the time, ensuring minimal downti...	QUALITY REQUIREMEN...
NFR04: Data should be encrypted using AES-256 to ensure security of user information.	QUALITY REQUIREMEN...
NFR05: Add a privacy policy page explaining how data is stored and used.	QUALITY REQUIREMEN...

REQUIREMENT MANAGEMENT TOOL

JIRA PROJECT CONSTRAINTS

2. Project Constraints

1. The LMS must be developed and deployed within 6 months.
2. The system must comply with GDPR and FERPA regulations for data privacy.
3. The LMS should be accessible on both mobile and desktop devices.
4. The system must be implemented using a scalable cloud-based architecture to support future growth.

NFR05: The LMS must be developed and deployed within 6 months.	PROJECT CONSTRAINTS
NFR06: The system must comply with GDPR and FERPA regulations for data privacy.	PROJECT CONSTRAINTS
NFR07: The LMS should be accessible on both mobile and desktop devices.	PROJECT CONSTRAINTS
NFR08: The system must be implemented using a scalable cloud-based architecture to support futur...	PROJECT CONSTRAINTS

Agile Process model

The **Agile process model** is an iterative approach that ensures flexibility, continuous improvement, and quick delivery. Instead of following a rigid sequence, development is divided into phases, where a working version of the **LMS** is released for feedback. This allows students, instructors, and administrators to test features early and suggest improvements. **Agile** enables quick adaptations to changing requirements. Regular reviews ensure quality, while cross-functional collaboration between developers, testers, and users enhances system efficiency. This makes **Agile** ideal for building a scalable and user-friendly **LMS**.

Why do we choose **Agile** process model?

1. Feedback

- In **Agile process model**, we get continuous feedback from students, instructors and admins after releasing a phase of **LMS** system. **Agile** allows incremental development, which means you can improve based on previous feedback.

2. Flexibility

- **LMS** may require changes or modifications. **Agile** allows easy modifications in case any requires from stakeholders.

3. Faster Delivery

- Instead of waiting until the system is complete, **Agile** allows you to develop and deploy core features first (e.g., Login, course management).

4. Risk management

- Regular testing in **Agile** makes us find bugs and errors early, which reduces overall project risk. If a feature isn't working properly after testing, you can change or modify it.

5. Better collaboration

- **Agile** helps developers, testers, and stakeholders work together smoothly through regular meetings, feedback, and updates.

JIRA EXPORT EXCEL FILE:

GITHUB LINK:

https://github.com/akhano014/ISE-Assignment/blob/main/se_lms_2025-03-11_11.05pm.csv

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